

COMP-SCI-5565

Intro to Statistical Learning

Topic 8 – Unsupervised Learning

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Lecture Objectives

- Compare and contrast supervised learning and unsupervised learning.
- Perform principal component analysis to analyze the sources of variance in a dataset.
- Perform K-means clustering to partition observations into a pre-specified number of clusters.
- Perform hierarchical clustering to partition observations into a tree-like structure.

Outline

8.1 Introduction

8.2 Principal Component Analysis

8.3 Clustering

8.4 K-Means

8.5 Hierarchical Clustering

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8.1 Introduction

Unsupervised vs Supervised Learning

- Most of this course focuses on supervised learning methods such as regression and classification.
- In that setting, we observe both a set of features $X_1, X_2, \dots, X_p$ for each object, as well as a response or outcome variable Y .
- The goal is then to predict Y using $X_1, X_2, \dots, X_p$.
- Here, we instead focus on unsupervised learning, we observe only the features $X_1, X_2, \dots, X_p$.
- We are not interested in prediction, because we do not have an associated response variable Y .

8.1 Introduction

Unsupervised vs supervised learning – continue

- The goal is to discover interesting things about the measurements: is there an informative way to visualize the data?
- Can we discover subgroups among the variables or the observations?
- We discuss two methods:
 - **Principal components analysis**: a tool used for data visualization or data pre-processing before supervised techniques are applied or dimensionality reduction.
 - **Clustering**: a broad class of methods for discovering unknown subgroups in data. E.g., K-means clustering and hierarchical clustering.

8.1 Introduction

Challenges of unsupervised learning

- Unsupervised learning is more subjective than supervised learning, as there is no simple goal for the analysis, such as predicting a response.
- Obtaining unlabeled data is often easier than labeling, which can require human intervention.
- For example, it is difficult to assess a movie review's overall sentiment automatically: is it favorable or not?
- But techniques for unsupervised learning are of growing importance in many fields:
 - subgroups of breast cancer patients grouped by their gene expression measurements,
 - groups of shoppers characterized by their browsing and purchase histories.

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8.2 Principal Component Analysis

- Suppose we have a dataset of n observations and p features, so we have a matrix $n \times p$.
- We want to select the best composition among our p features to achieve the highest value of data representation in a lower dimension.
- Another scope of the unsupervised principal component analysis is data visualization.
- To obtain a visualization of all the components of a dataset in a lower dimension space, and have an idea of their relations.

8.2 Principal Component Analysis

Steps to PCA

(1) Variables normalization

$$z = \frac{x - \bar{x}}{sd(x)}$$

where

$\bar{x}$ = mean

sd = standard deviation of x .

- PCA is affected by scale, so the data needs to be normalized.
- Given $n \times p$, the normalized data has the same dimension $n \times p$.

8.2 Principal Component Analysis

Steps to PCA – continued

(2) Calculating the covariance matrix

- To understand how the input variables vary from the mean to each other.

$$\text{cov}(x, y) = \frac{\sum_1^n (x_i - \bar{x})(y_i - \bar{y})}{n - 1}$$

- The general form of the covariance matrix is as follows

$$\begin{bmatrix} \text{Var}(x_1) & \cdots & \text{Cov}(x_1, x_n) \\ \vdots & \cdot & \vdots \\ \vdots & & \vdots \\ \text{Cov}(x_n, x_1) & \cdots & \text{Var}(x_n) \end{bmatrix}$$

- Given $n \times p$, the covariance output is a square matrix with dimension $p \times p$.

8.2 Principal Component Analysis

Steps to PCA – continued

(3) Computing eigenvectors and eigenvalues of the covariance matrix

- For a given covariance matrix A , a vector v is an eigenvector of A if the following equation is satisfied:

$$Av = \lambda v$$

- Here, λ represents the eigenvalue associated with the eigenvector v .
- To find λ and v , we solve the characteristic equation $\det(A - \lambda I) = 0$, where I is the identity matrix, and $\det$ represents the determinant of a matrix.
- Eigenvectors represent the principal components, with dimension $p \times p$.
- Eigenvalues represent the **proportion of variance explained** by each component, with dimension $1 \times p$.

8.2 Principal Component Analysis

Steps to PCA – continued

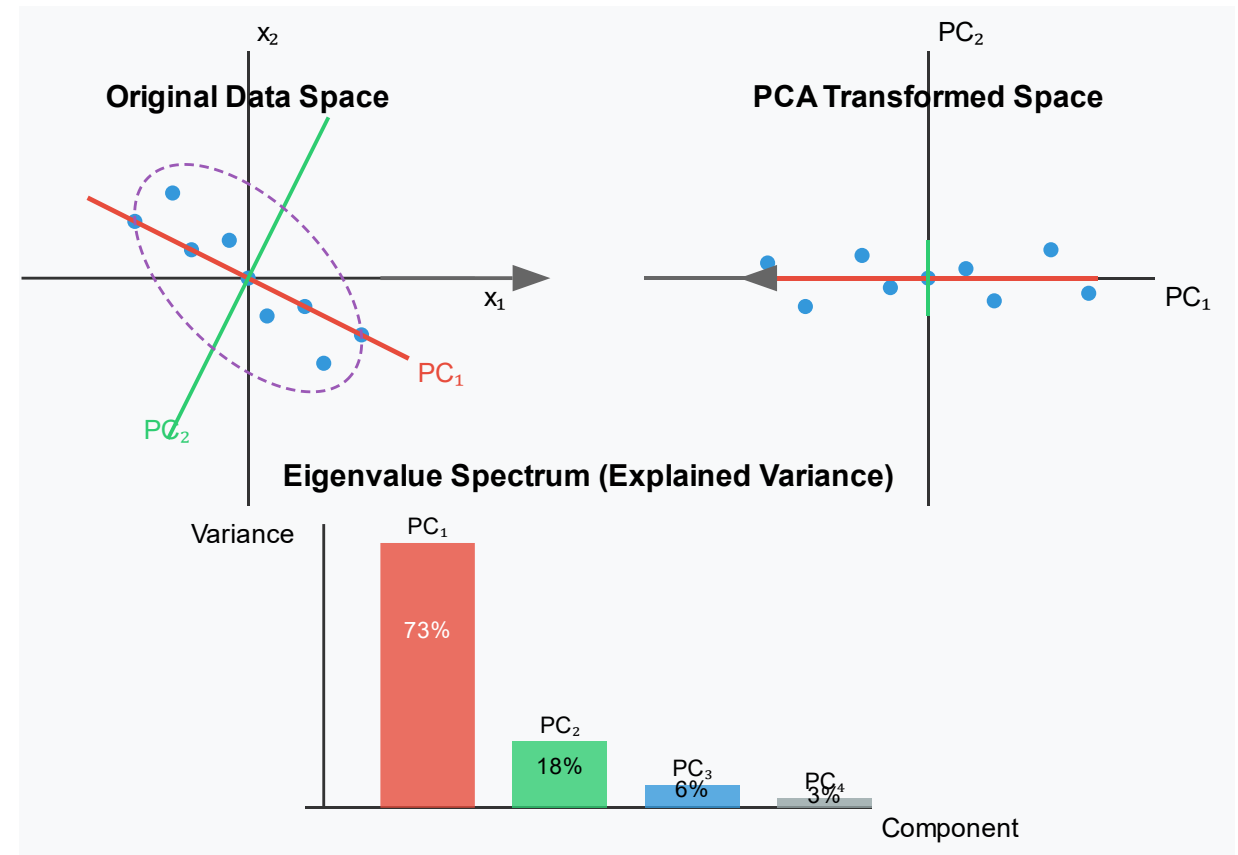
(3) Computing eigenvectors and eigenvalues of the covariance matrix

• Original Data Space

- Blue dots represent data points in the original feature space (x_1, x_2)
- The purple ellipse represents the covariance structure
- Redline (PC_1) is the first eigenvector, pointing in the direction of maximum variance
- The green line (PC_2) is the second eigenvector, orthogonal to PC_1

• PCA Transformed Space

- The same data points after projection onto the eigenvectors
- The coordinate system has been rotated to align with the directions of maximum variance.
- Note how the variance is maximized along PC_1 and minimized along PC_2



8.2 Principal Component Analysis

Steps to PCA – continued

(4) Choosing components and forming a feature vector

- By ranking the eigenvalues in descending order and choosing the top k eigenvectors.
- We set $k < p$.

(5) Deriving the new data set

- This is done by multiplying the original data set by the feature vector (eigenvectors).

$$PCA = \text{normalized } X_{n \times p} \times \text{eigenvectors}_{p \times k}$$

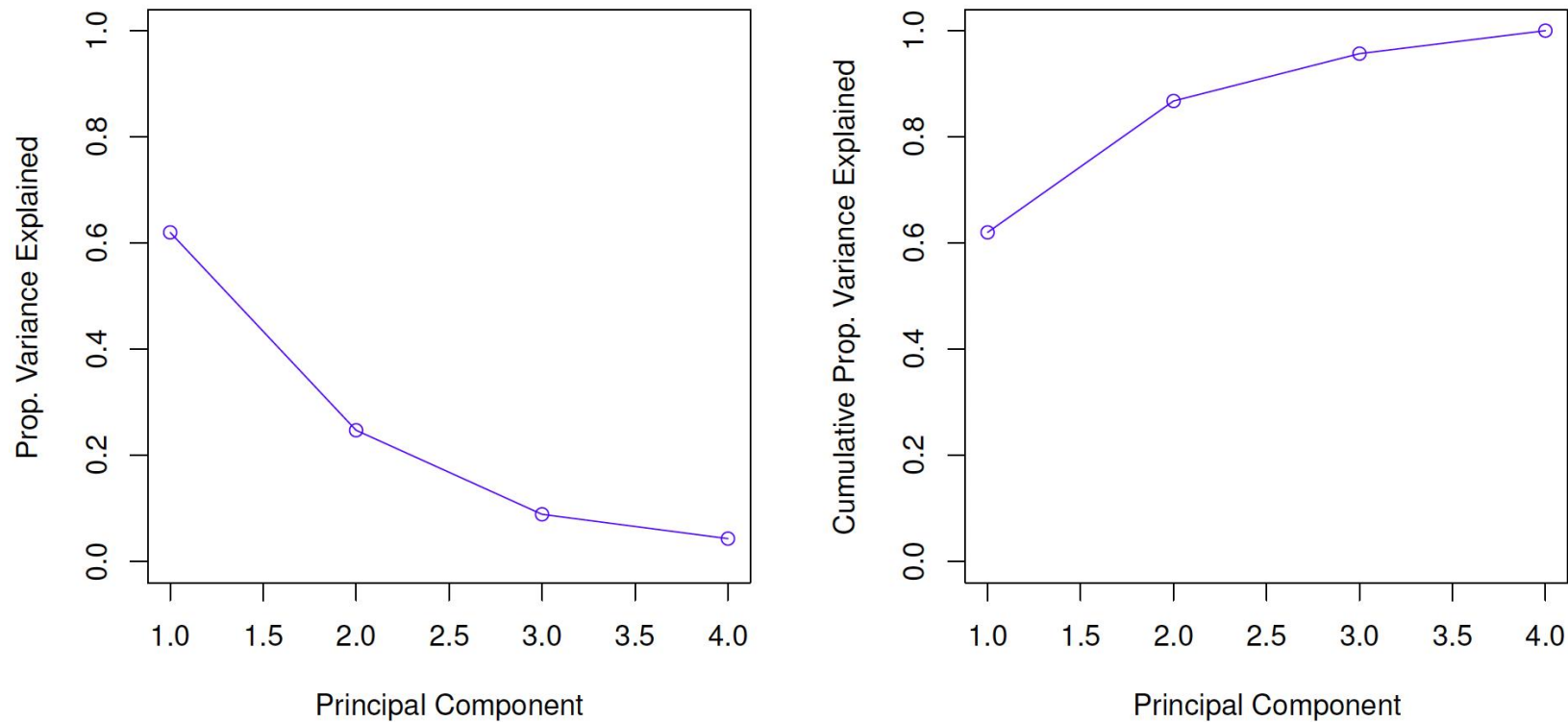
8.2 Principal Component Analysis

Choosing the number of principal components

- The number of principal components retained in the analysis is a critical decision.
- There is no simple answer to this question, as cross-validation is not available. Why not?
- In practice, the choice is often based on the **cumulative explained variance**, which should be as high as possible while keeping fewer components.
- A common rule of thumb is to keep the principal components that explain at least 85% of the variance.

8.2 Principal Component Analysis

Choosing the number of principal components – continued



The “scree plot” can be used as a guide: we look for an “elbow”.

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8.3 Clustering

- Clustering refers to a broad set of techniques for finding subgroups or clusters in a data set.
- We seek a partition of the data into distinct groups so that the observations within each group are quite similar to each other,
- To make this concrete, we must define what it means for two or more observations to be similar or different.
- Indeed, this is often a domain-specific consideration that must be made based on knowledge of the data being studied.



8.3 Clustering

What is the difference between PCA and Clustering?

- PCA looks for a low-dimensional representation of the observations that explains a good fraction of the variance.
- Clustering looks for homogeneous subgroups among the observations.

8.3 Clustering

Clustering for market segmentation

- Suppose we have access to many measurements (e.g., median household income, occupation, distance from nearest urban area, and so forth) for many people.
- We aim to perform **market segmentation** by identifying subgroups of people who might be more receptive to a particular form of advertising or more likely to purchase a particular product.
- The task of performing market segmentation amounts to clustering the people in the data set.

8.3 Clustering

Two clustering methods

K-means

- We seek to partition the observations into a pre-specified number of clusters in K-means clustering.

Hierarchical clustering

- In hierarchical clustering, we do not know how many clusters we want.
- We end up with a tree-like visual representation of the observations called a dendrogram.
- That allows us to view at once the clustering obtained for each possible number of clusters, from 1 to n .

Outline

8.1 Introduction

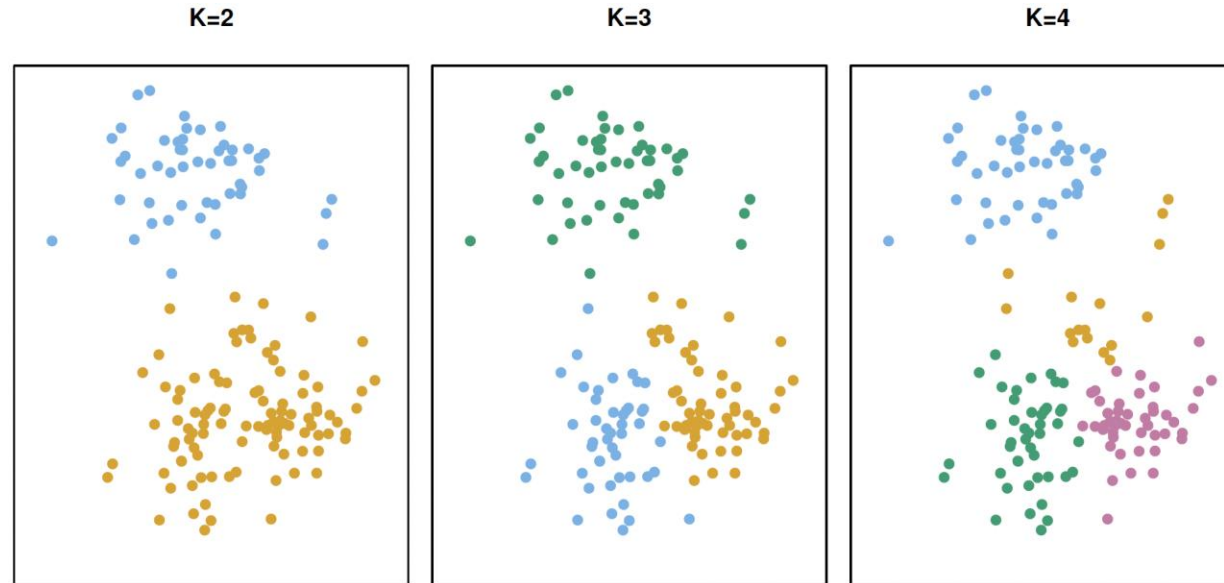
8.2 Principal Component Analysis

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8.4 K-Means



- We seek to partition the observations into a pre-specified clustering number of clusters.

Steps to K-means:

1. Decide the number of clusters K
2. Then the K-means algorithm will assign each observation to exactly one of the K clusters

8.4 K-Means

- Let $C_1, \dots, C_k$ denote sets containing the indices of the observations in each cluster.
- A good clustering is one for which the **within-cluster variation** is as small as possible.
- Within-cluster variation measure: $W(C_k)$. Hence, we want to solve the problem

$$\min_{C_1, \dots, C_k} \left\{ \sum_{k=1}^K W(C_k) \right\}$$

- In words, this formula says that we want to partition the observations into K clusters such that the total within-cluster variation, summed over all K clusters, is as small as possible.
- Typically we use **Euclidean distance**.

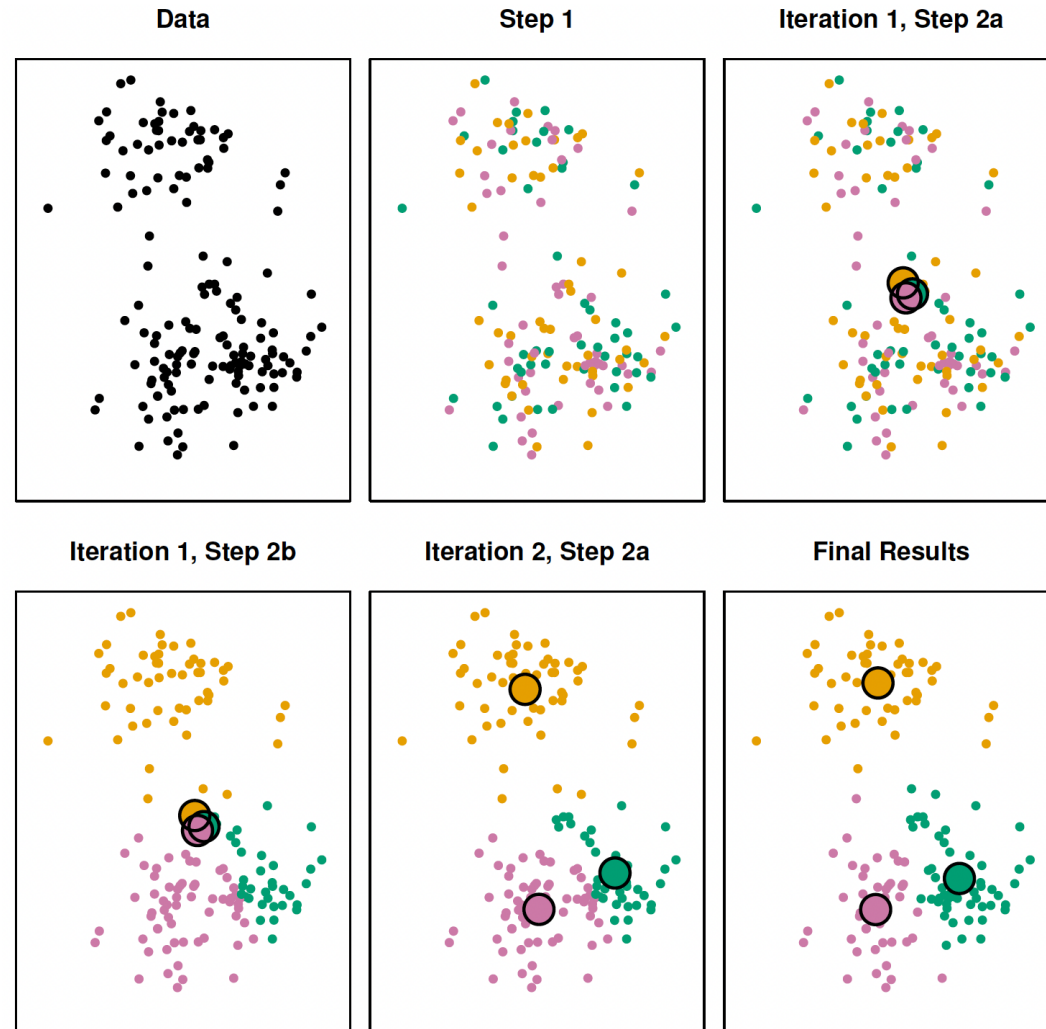
8.4 K-Means

K-Means clustering algorithm

1. Randomly assign a number, from 1 to K , to each observation. These serve as initial cluster assignments for the observations.
2. Iterate until the cluster assignments stop changing:
 - 2.1 For each of the K clusters, compute the cluster centroid.
The k th cluster centroid is the vector of the p feature means for the observations in the k th cluster.
 - 2.2 Assign each observation to the cluster whose centroid is closest (where *closest* is determined using Euclidean distance).

8.4 K-Means

Example



8.4 K-Means

Example – continued

Here are the details of the previous figure

The progress of the K-means algorithm with $K=3$.

- *Top left:* The observations are shown.
- *Top center:* In Step 1 of the algorithm, each observation is randomly assigned to a cluster.
- *Top right:* In Step 2(a), the cluster centroids are computed. These are shown as large colored disks. Initially, the centroids are almost completely overlapping because the initial cluster assignments were randomly chosen.
- *Bottom left:* In Step 2(b), each observation is assigned to the nearest centroid.
- *Bottom center:* Step 2(a) is again performed, leading to new cluster centroids.
- *Bottom right:* The results obtained after 10 iterations.

8.4 K-Means

Example with different starting values



8.4 K-Means

Example with different starting values – continued

Here are the details of the previous figure

- K-means clustering was performed six times on the data from the previous figure with $K = 3$, each with a different random assignment of the observations in Step 1 of the K-means algorithm.
- Above each plot is the value of the objective – $W(C_k)$.
- Three different local optima were obtained, one of which resulted in a smaller value of the objective and provided better separation between the clusters.
- Those labeled in red all achieved the same best solution, with an objective value of 235.8

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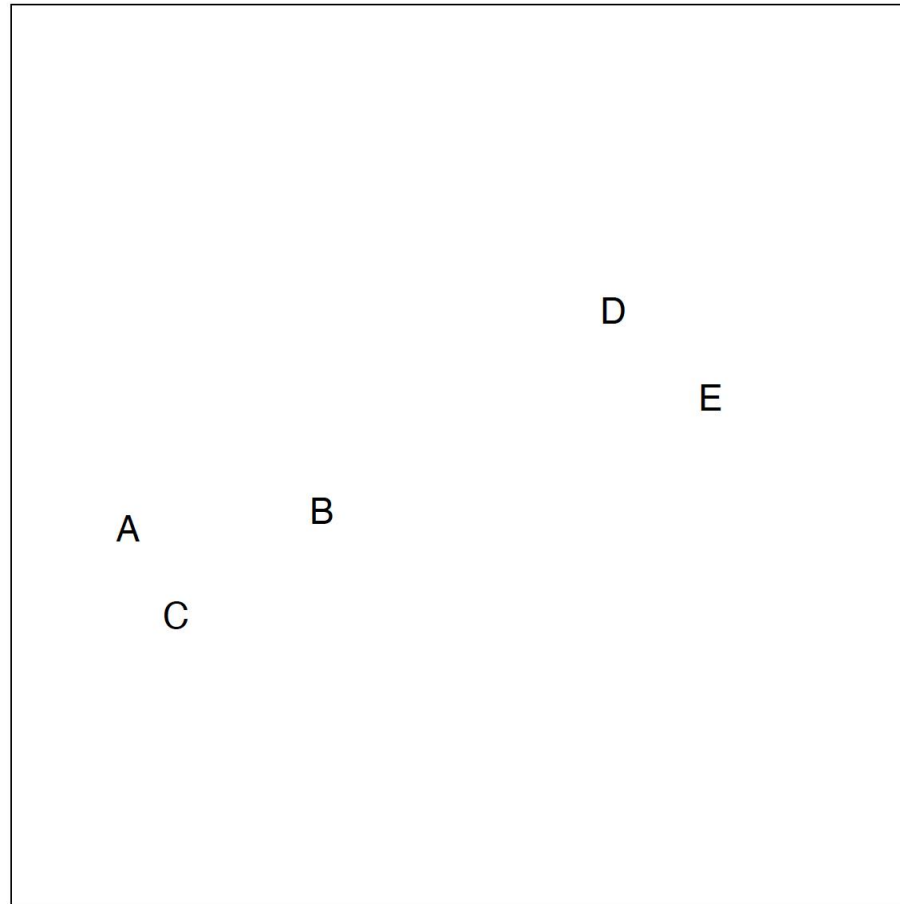
8.5 Hierarchical Clustering

- K-means clustering requires us to pre-specify the number of clusters K .
- This can be a disadvantage (cross-validation can be used to choose K).
- **Hierarchical clustering** is an alternative approach that does not require committing to a particular choice of K .
- In this section, we describe bottom-up or agglomerative clustering.
- This is the most common type of hierarchical clustering and refers to the fact that a dendrogram is built starting from the leaves and combining clusters up to the trunk.

8.5 Hierarchical Clustering

The idea

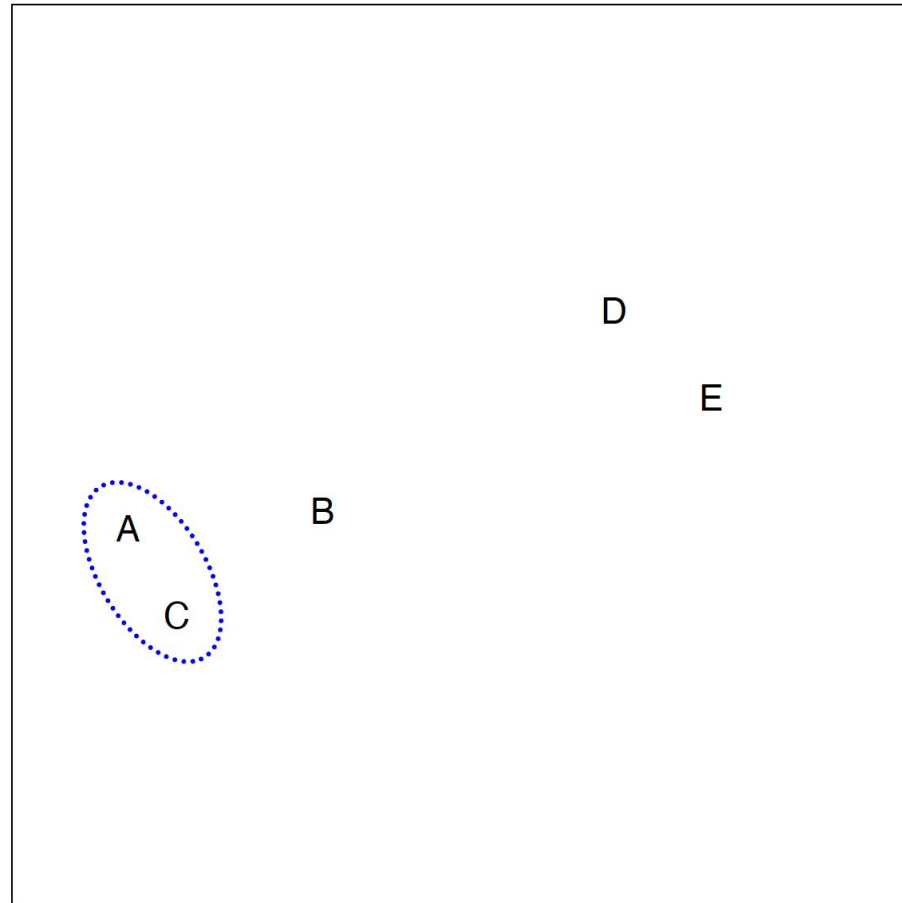
- Builds a hierarchy in a “bottom-up” fashion...



8.5 Hierarchical Clustering

The idea – continued

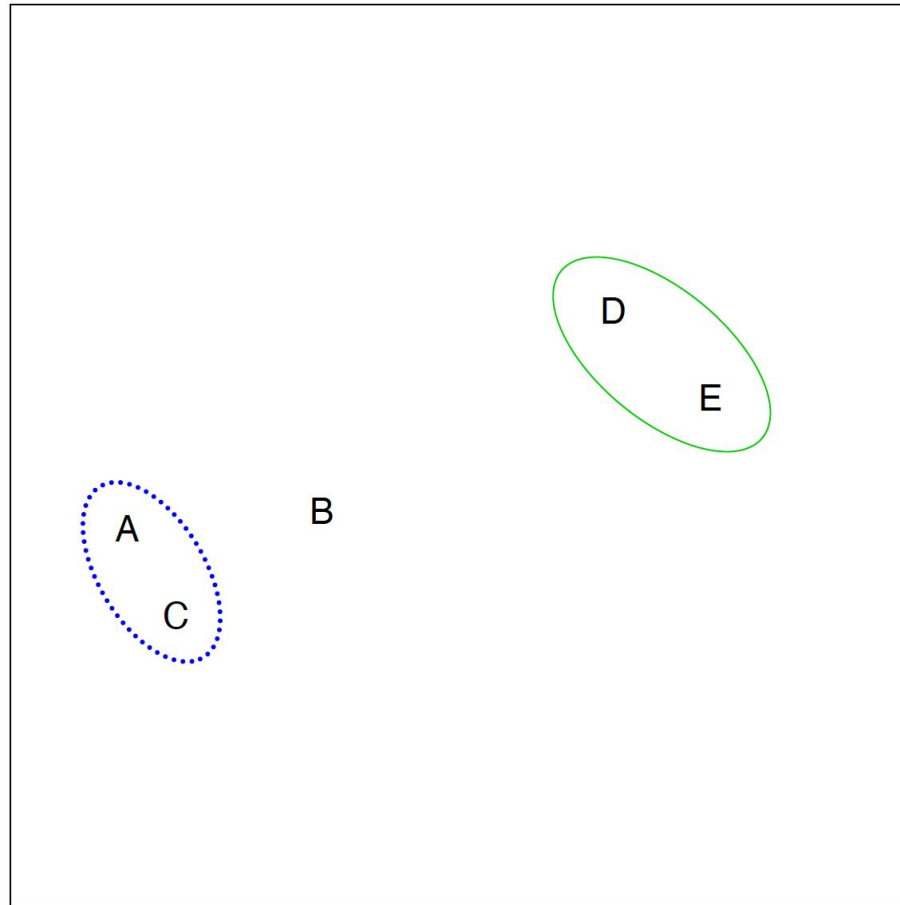
- Builds a hierarchy in a “bottom-up” fashion...



8.5 Hierarchical Clustering

The idea – continued

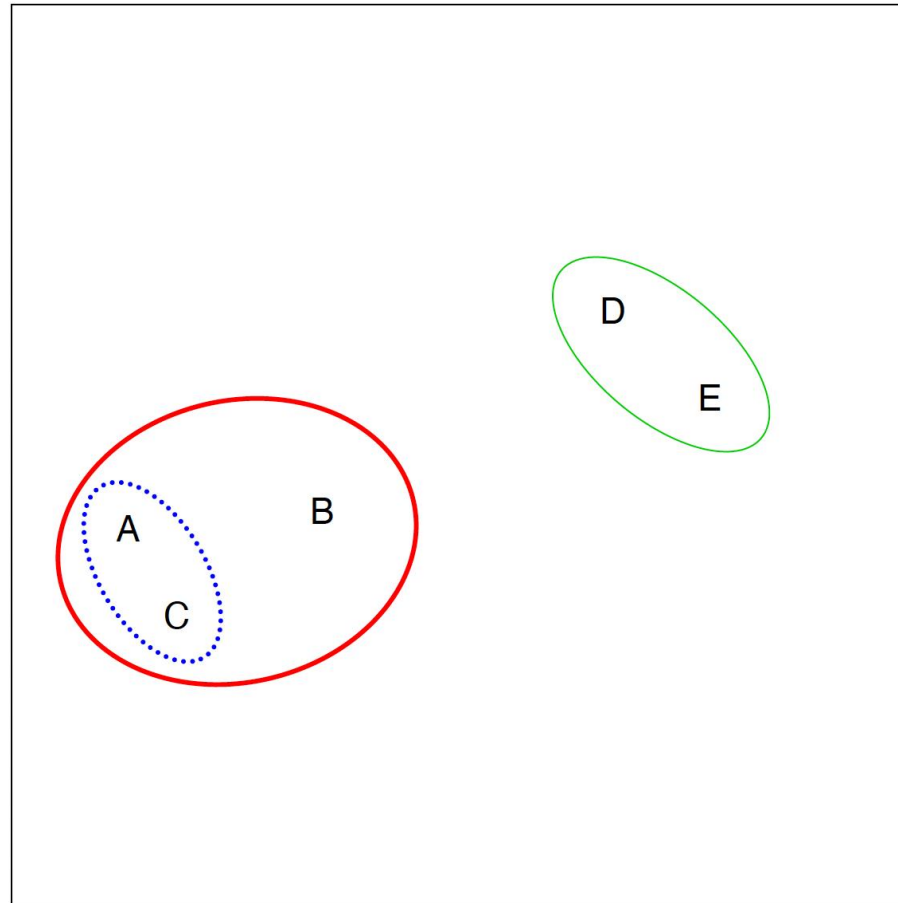
- Builds a hierarchy in a “bottom-up” fashion...



8.5 Hierarchical Clustering

The idea – continued

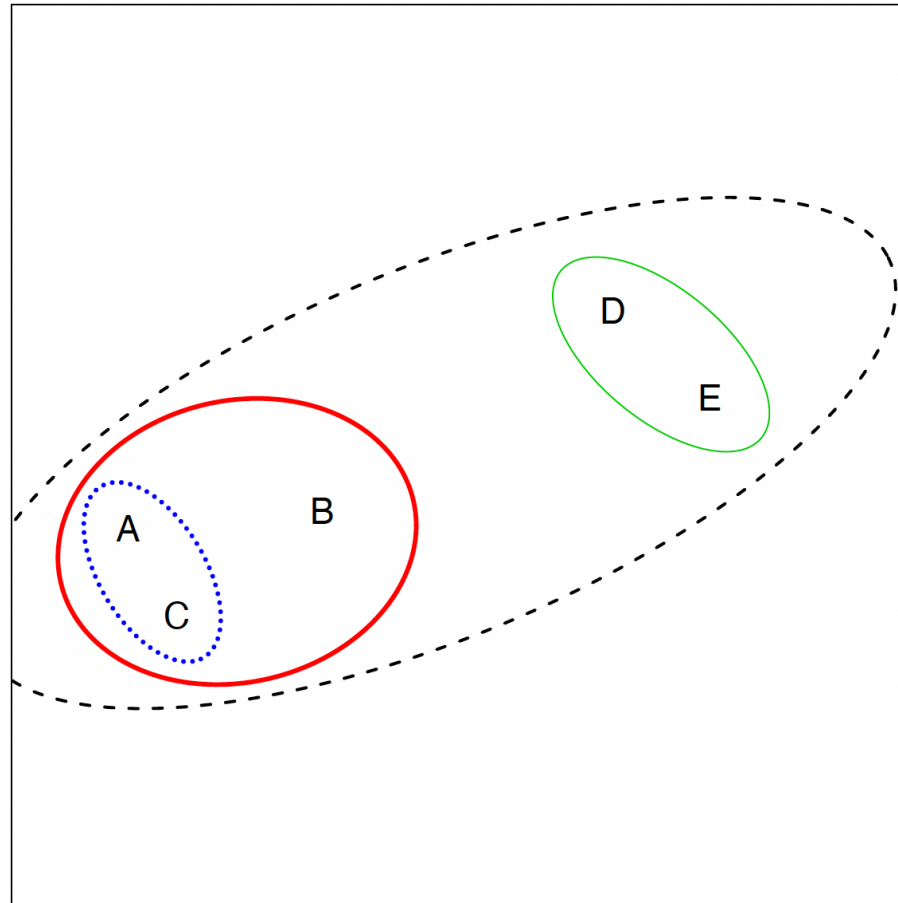
- Builds a hierarchy in a “bottom-up” fashion...



8.5 Hierarchical Clustering

The idea – continued

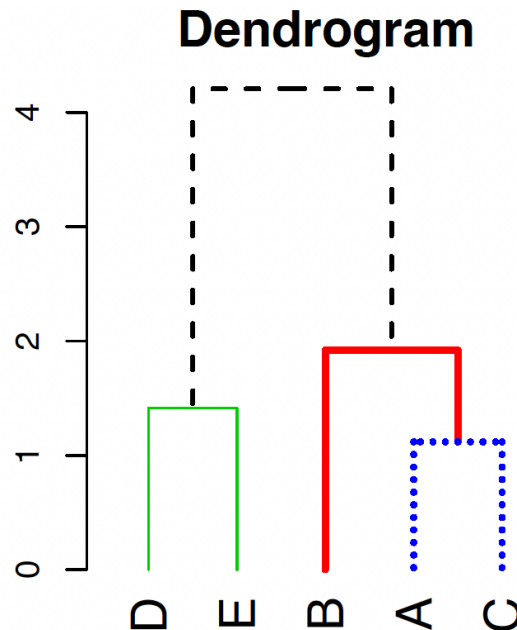
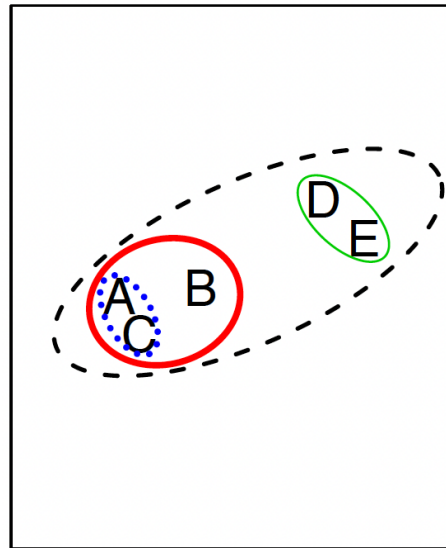
- Builds a hierarchy in a “bottom-up” fashion...



8.5 Hierarchical Clustering

Hierarchical clustering algorithm

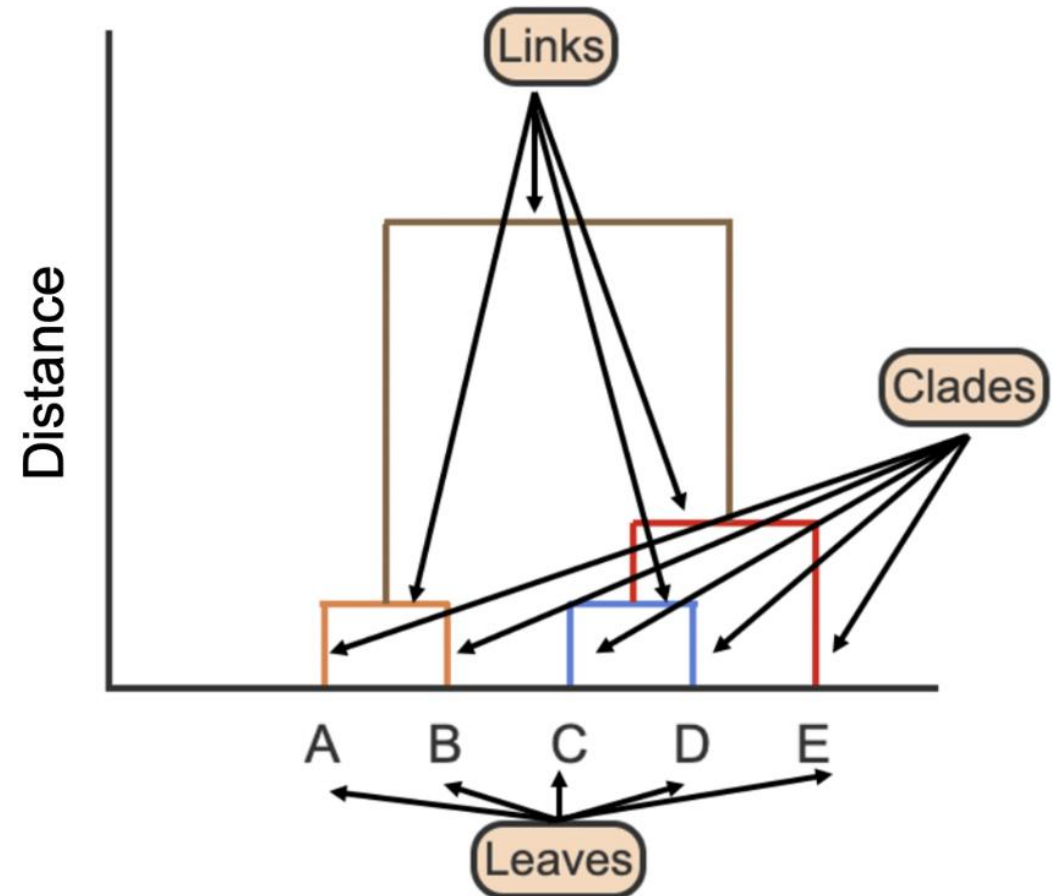
1. Start with each point in its own cluster.
2. Identify the **closest** two clusters and merge them.
3. Repeat.
4. Ends when all points are in a single cluster.



8.5 Hierarchical Clustering

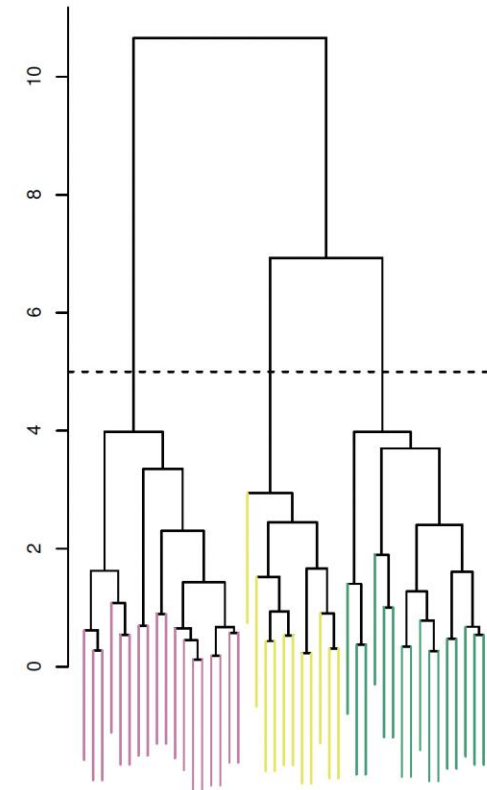
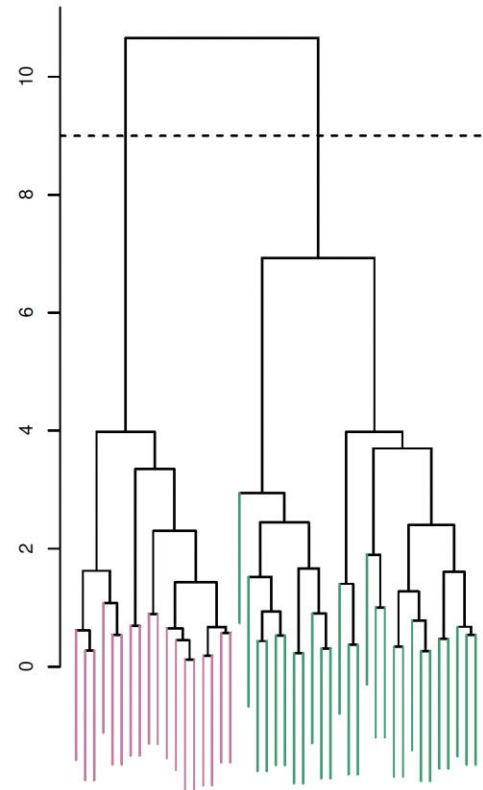
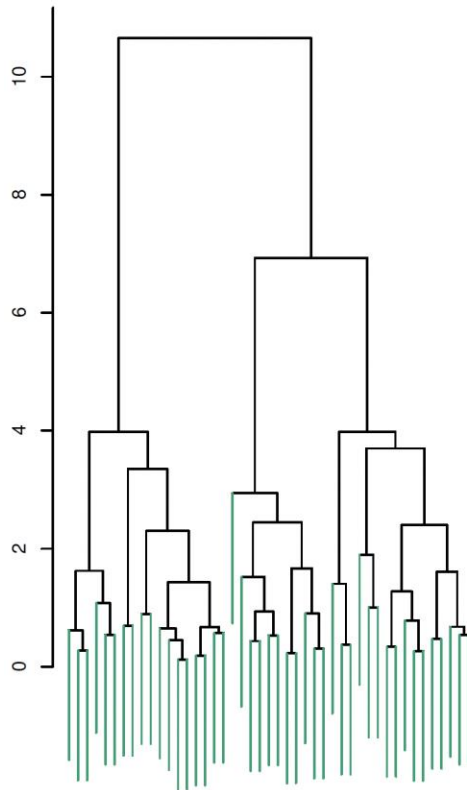
Dendrograms

- The output of a hierarchical clustering algorithm is a *dendrogram*.
- A **dendrogram** is a tree that shows the order in which clusters are grouped together and the distances between clusters. Parts of a dendrogram are listed below:
 - A **clade** is a branch of a dendrogram or a vertical line.
 - A **link** is a horizontal line that connects two clades, whose height gives the distance between clusters.
 - A **leaf** is the terminal end of each clade in a dendrogram, which represents a single instance.



8.5 Hierarchical Clustering

Example



8.5 Hierarchical Clustering

Example – continued

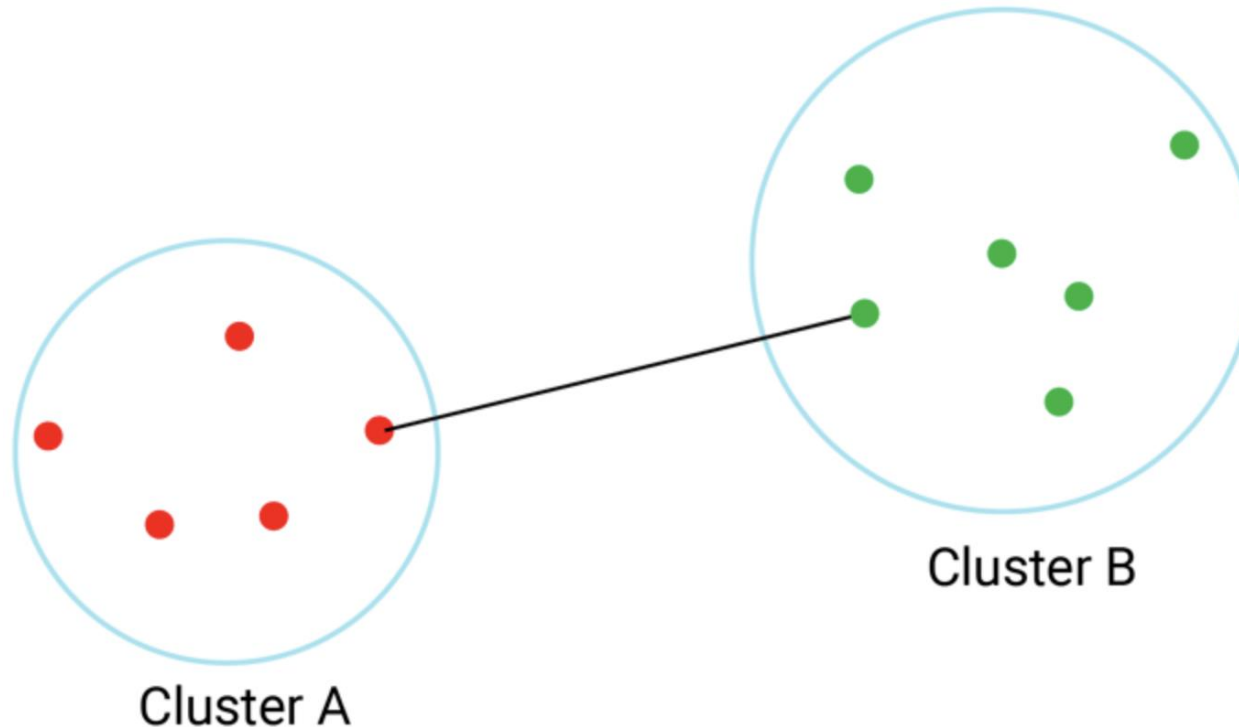
- *Left*: Dendrogram obtained from hierarchically clustering the data from the previous slide, with complete linkage and Euclidean distance.
- *Center*: The dendrogram from the left-hand panel is cut at 9 (indicated by the dashed line). This cut results in two distinct clusters, shown in different colors.
- *Right*: The dendrogram from the left-hand panel is now cut at 5. This cut results in three distinct clusters, shown in different colors.
- Note that the colors were not used in clustering but simply for display purposes in this figure.
- The **height** of the blocks represents the distance between clusters.
- Distance between observations represents **dissimilarities**.

8.5 Hierarchical Clustering

Types of linkage

Single: Minimal inter-cluster dissimilarity.

- Compute all pairwise dissimilarities between the observations in cluster A and the observations in cluster B, and record the smallest of these dissimilarities.

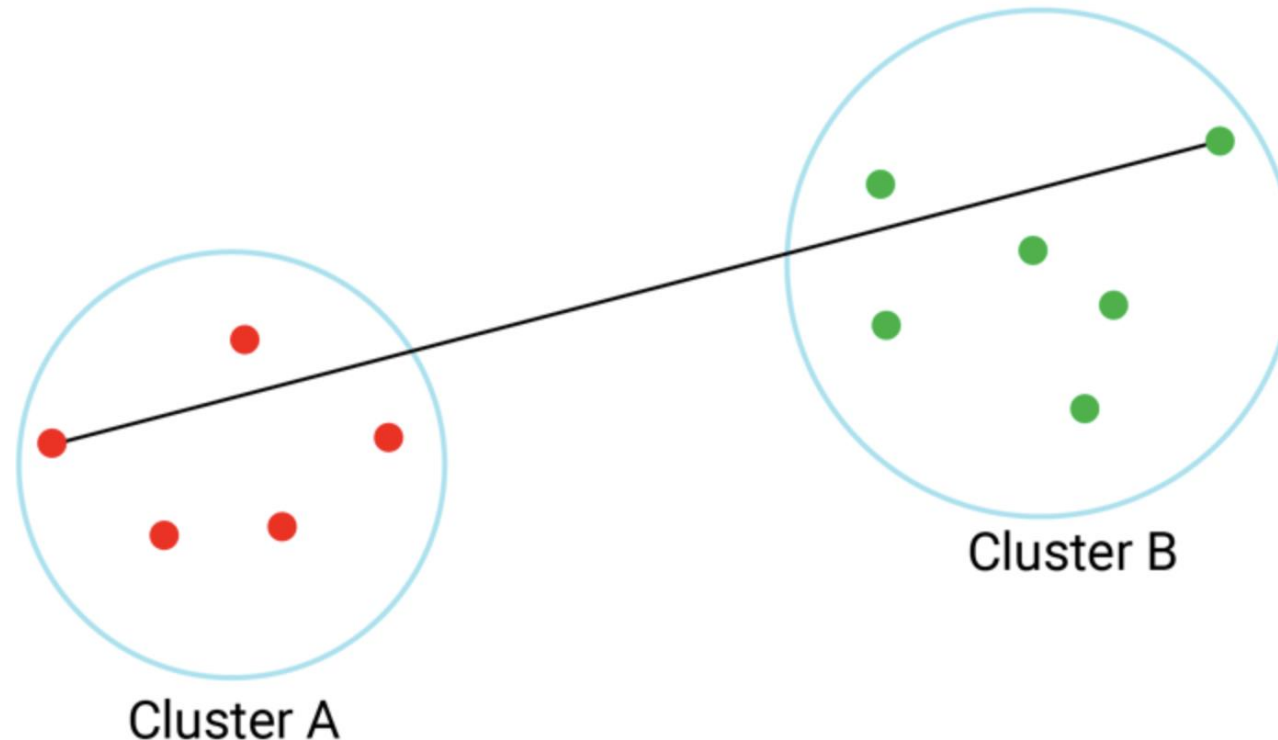


8.5 Hierarchical Clustering

Types of linkage

Complete: Maximal inter-cluster dissimilarity.

- Compute all pairwise dissimilarities between the observations in cluster A and the observations in cluster B, and record the largest of these dissimilarities.

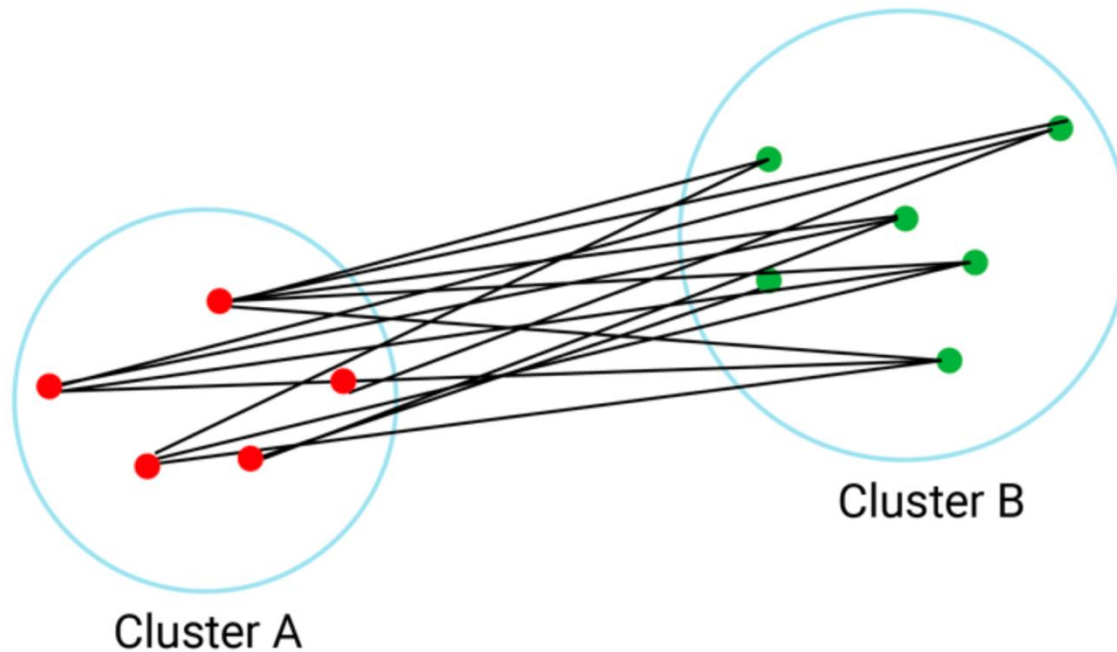


8.5 Hierarchical Clustering

Types of linkage

Average: Mean inter-cluster dissimilarity

- Compute all pairwise dissimilarities between the observations in cluster A and the observations in cluster B, and record the average of these dissimilarities.
- Less affected by outliers.

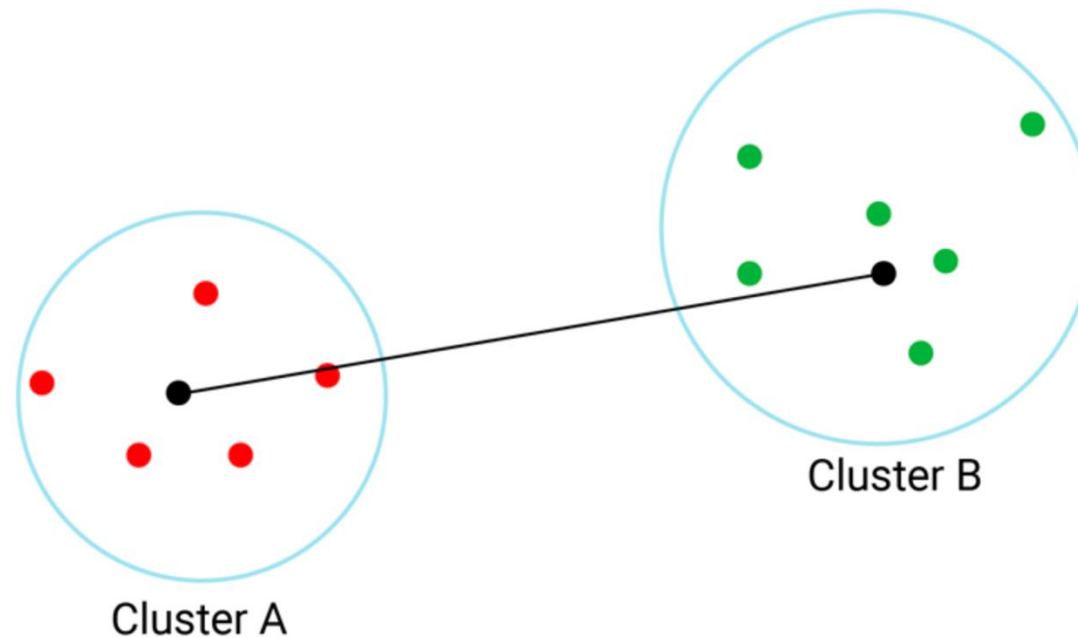


8.5 Hierarchical Clustering

Types of linkage

Centroid

- Dissimilarity between the centroid for cluster A and cluster B.

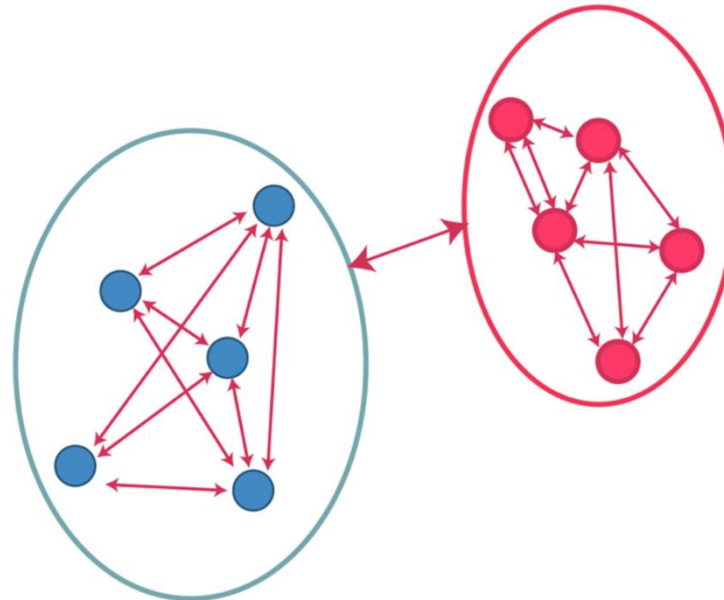


8.5 Hierarchical Clustering

Types of linkage

Ward's-linkage

- Ward linkage is the distance that minimizes variance in the cluster and maximizes variance between the clusters.
- Consider merging two clusters; based on how it changes the total distance from centroids.



8.5 Hierarchical Clustering

What type of linkage should be used?

- Average and complete linkage are the two most popular distance metrics in hierarchical clustering.
- However, when there are no clear theoretical justifications for the choice of linkage criteria, Ward's method is the default option.

Summary

- *Unsupervised learning* is important for understanding the variation and grouping structure of a set of unlabeled data and can be a useful pre-processor for supervised learning.
- It is intrinsically more difficult than supervised learning because there is no gold standard (like an outcome variable) and no single objective (like test set accuracy).
- It is an active field of research, with many recently developed tools, such as *self-supervised learning*.

Further Reading

- Chapter 12 – James G., Witten D., Hastie T., Tibshirani R., & Taylor J. (2023) *An Introduction to Statistical Learning with Applications in Python*. Springer
<https://link.springer.com/book/10.1007/978-3-031-38747-0>
- Chapter 12 - Introduction to Statistical Learning Using R Book Club:
<https://r4ds.github.io/bookclub-islr/support-vector-machines.html>
- Summary: <https://www.bijenpatel.com/guide/islr/unsupervised-learning/>
- PCA: <https://deepai.org/machine-learning-glossary-and-terms/principal-components-analysis>
- Hierarchical Clustering: <https://ibkrcampus.com/ibkr-quant-news/hierarchical-clustering-in-python-a-comprehensive-implementation-guide-part-ii/>

End of Topic

Thank you
Any questions?